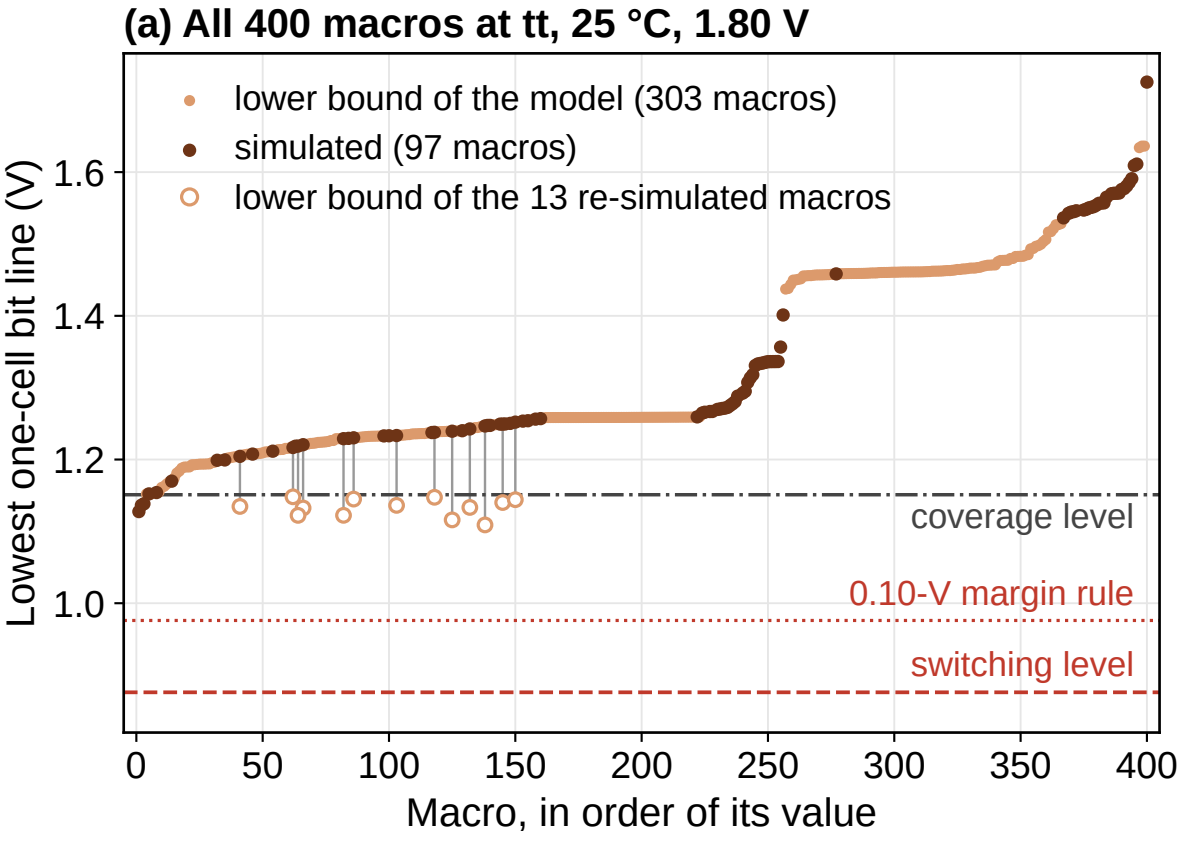
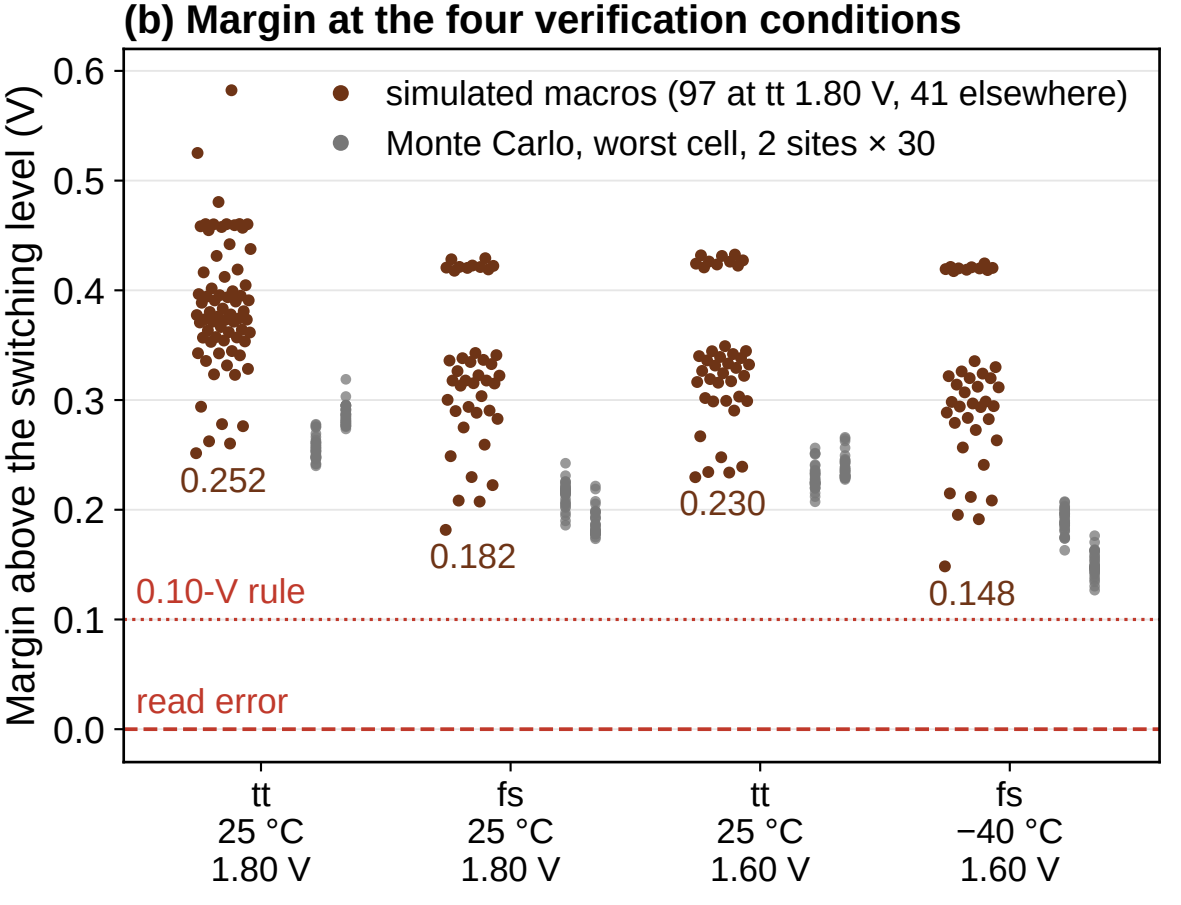

(a) All 400 macros at tt, 25 °C, 1.80 V
lower bound of the model (303 macros)
simulated (97 macros)
lower bound of the 13 re-simulated macros
Lowest one-cell bit line (V)
1.6
1.4
1.2
1.0
coverage level
0.10-V margin rule
switching level
0
50
100
150
200
250
300
350
400
Macro, in order of its value


(b) Margin at the four verification conditions
simulated macros (97 at tt 1.80 V, 41 elsewhere)
Monte Carlo, worst cell, 2 sites × 30
Margin above the switching level (V)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.252
0.182
0.230
0.148
0.10-V rule
read error
tt
25 °C
1.80 V
fs
25 °C
1.80 V
tt
25 °C
1.60 V
fs
−40 °C
1.60 V